The leading constant for $S_n \times A$

Let $n \in \{3,4,5\}$, and let A be a finite abelian group of order $m = |A|$. Then we consider $G = S_n \times A$. Fix a signature (R,S) with $R + 2S = nm$. Write $N_G(X)$ for the number of $\mathbb{Q}$-isomorphism classes of degree mn fields L with signature (R,S), $\mathrm{Gal}(L^c/\mathbb{Q}) \simeq G$, and $|\operatorname{Disc} L| < X$. The constant in $N_G(X) \sim C_G X^{1/m}$ as $X \to \infty$ is given by

$$C_G := \frac{1}{|\operatorname{Aut} A|} \sum_{\substack{B \le A \\ \chi \in \mathrm{Hom}(B, \mathbb{C}^\times)}} \frac{\mu(B,A)}{|B|} \left[\frac{1}{2} \sum_{\substack{r+2s=n \\ mr=R}} \frac{1}{r!\, 2^s s!} + \mathbf{1}_{\{R=0\}} \kappa_n \big(|B| \mathbf{1}_{\{\chi = \mathbf{1}\}} - 1 \big) \right]$$
$$\times \prod_{p \text{ prime}} \left[\left(1 - \frac{1}{p}\right) \sum_{\psi \in \mathrm{Hom}_{\mathrm{cont}}(\mathbb{Z}_p^\times, B)} \chi(\psi(-1)) \sum_{E \in \mathcal{E}_n(p)} \frac{p^{-d_p(E \otimes_{\mathbb{Q}_p} L_\psi)/h_\psi}}{|\operatorname{Aut}_{\mathbb{Q}_p}(E)|} \right].$$

Notation. The constants are $\kappa_3 = 1/3$, $\kappa_4 = 5/24$ and $\kappa_5 = 13/120$. The sum is over all subgroups B of A. The Möbius function is defined by

$$\mu(B,A) := \begin{cases} \prod_p (-1)^{r_p} p^{\frac{r_p(r_p-1)}{2}} & \text{if } A/B \cong \prod_p (\mathbb{Z}/p\mathbb{Z})^{r_p}, \\ 0, & \text{otherwise.} \end{cases}$$

For $\psi \in \mathrm{Hom}_{\mathrm{cont}}(\mathbb{Z}_p^\times, B)$, extend ψ to $\mathbb{Q}_p^\times$ by $\psi(p) = 1$, and let $L_\psi/\mathbb{Q}_p$ be the abelian extension associated to its kernel by local class field theory. Moreover, we let $h_\psi = [L_\psi : \mathbb{Q}_p] = |\operatorname{im} \psi|$.

Also, we denote $\mathcal{E}_n(p)$ as the finite set of isomorphism classes of degree n étale $\mathbb{Q}_p$-algebras. For $E \in \mathcal{E}_n(p)$, let $d_p(E \otimes_{\mathbb{Q}_p} L_\psi)$ be the discriminant exponent of the maximal $\mathbb{Z}_p$-order in $E \otimes_{\mathbb{Q}_p} L_\psi$. Finally, $\mathrm{Aut}_{\mathbb{Q}_p}(E)$ is the group of $\mathbb{Q}_p$-algebra automorphisms of E.

Now, we give some concrete examples and numerics (all signatures are included):

Some numerics

n	$C_{S_n \times C_2}$	$C_{S_n \times (C_2 \times C_2)}$
3	0.600485169929...	0.685009888568...
4	0.274102386235...	0.131474579489...
5	0.067505846256...	0.012294591845...

The asymptotic powers are $X^{1/2}$ and $X^{1/4}$ respectively.

The cubic cases

$$C_{S_3 \times C_2} = \frac{1}{3} \left[\frac{68 + 7\sqrt{2}}{64} \prod_{p>2} \left(1 + \frac{2}{p^{3/2}} - \frac{1}{p^{5/2}} - \frac{1}{p^3} - \frac{1}{p^{7/2}}\right) - \frac{7}{8} \prod_{p>2} \left(1 - \frac{1}{p^3}\right) \right].$$

$$C_{S_3\times(C_2\times C_2)} = \frac{1}{18}\Bigg[\frac{208+45\sqrt{2}}{128}\prod_{p>2}\left(1+\frac{6}{p^{3/2}}-\frac{3}{p^{5/2}}-\frac{1}{p^3}-\frac{3}{p^{7/2}}\right) - \frac{3(68+7\sqrt{2})}{64}\prod_{p>2}\left(1+\frac{2}{p^{3/2}}-\frac{1}{p^{5/2}}-\frac{1}{p^3}-\frac{1}{p^{7/2}}\right) + \frac{7}{4}\prod_{p>2}\left(1-\frac{1}{p^3}\right)\Bigg].$$

The quartic cases

$$C_{S_4\times C_2} = \frac{5}{24}\left[\frac{353}{256}\prod_{p>2}\left(1+\frac{4}{p^2}-\frac{2}{p^3}-\frac{3}{p^4}\right) - \frac{17}{16}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^3}-\frac{1}{p^4}\right)\right].$$

$$C_{S_4\times(C_2\times C_2)} = \frac{5}{144}\Bigg[\frac{560+3\sqrt{2}}{256}\prod_{p>2}\left(1+\frac{10}{p^2}-\frac{4}{p^3}-\frac{7}{p^4}\right) - \frac{1059}{256}\prod_{p>2}\left(1+\frac{4}{p^2}-\frac{2}{p^3}-\frac{3}{p^4}\right) + \frac{17}{8}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^3}-\frac{1}{p^4}\right)\Bigg].$$

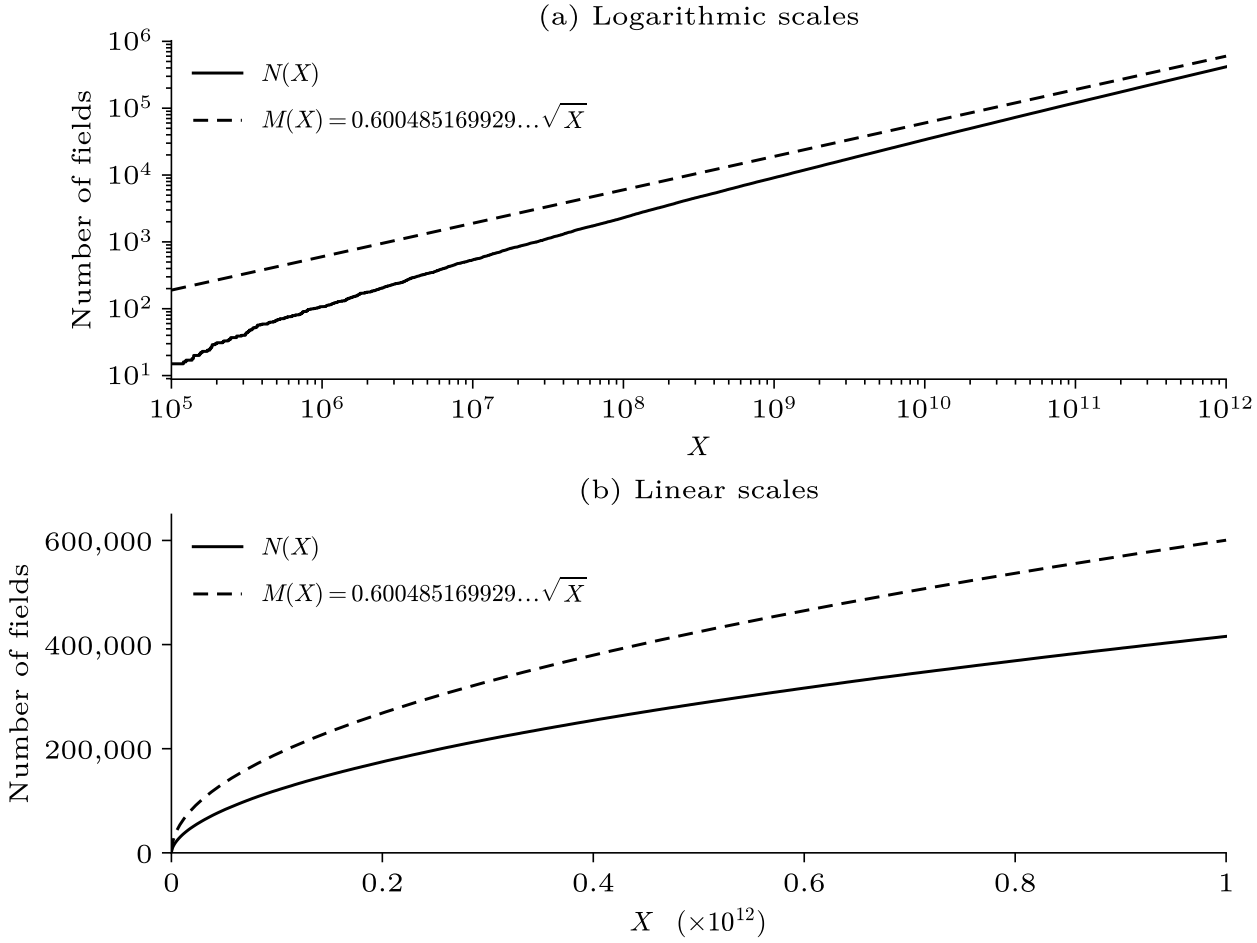
The quintic cases

$$C_{S_5\times C_2} = \frac{13}{120}\left[\frac{648+19\sqrt{2}}{512}\prod_{p>2}\left(1+\frac{1}{p^2}+\frac{3}{p^{5/2}}-\frac{1}{p^4}-\frac{2}{p^{9/2}}-\frac{1}{p^5}-\frac{1}{p^{11/2}}\right) - \frac{37}{32}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^4}-\frac{1}{p^5}\right)\right].$$

$$C_{S_5\times(C_2\times C_2)} = \frac{13}{720}\Bigg[\frac{3133+243\sqrt{2}}{2048}\prod_{p>2}\left(1+\frac{1}{p^2}+\frac{9}{p^{5/2}}-\frac{1}{p^4}-\frac{6}{p^{9/2}}-\frac{1}{p^5}-\frac{3}{p^{11/2}}\right) - \frac{3(648+19\sqrt{2})}{512}\prod_{p>2}\left(1+\frac{1}{p^2}+\frac{3}{p^{5/2}}-\frac{1}{p^4}-\frac{2}{p^{9/2}}-\frac{1}{p^5}-\frac{1}{p^{11/2}}\right) + \frac{37}{16}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^4}-\frac{1}{p^5}\right)\Bigg].$$

The leading constant for $S_n \times A$

For $G = S_3 \times C_2$ ($\simeq D_6$), the two functions below are the actual count $N(X) = N_{S_3\times C_2}(X)$ and its leading term $M(X) = C_{S_3\times C_2}\sqrt{X} = (0.600485169929\ldots)\sqrt{X}$. The solid curve represents $N(X)$ and the dashed curve represents $M(X)$.



Some numerics

X	$N(X)$	$M(X)$
10^6	108	600.48...
10^8	2299	6004.85...
10^{10}	33770	60048.51...
10^{11}	120217	189890.08...
10^{12}	415740	600485.16...

The counts include all signatures and are complete for $X \le 10^{12}$ by using the PARI/GP cubic tables and all relevant quadratic fields. Their discriminant and signature multiplicities below 10^7 agree with the LMFDB.